

Professional Database Normalization Walkthrough

Based on text infographic for e-commerce 'Orders' table

✓ Starting point – Unnormalized Table (UNF)

Orders_UNF					
order_id	cust_name	cust_email	products		address
			name	qty price	
1			(Mouse,2,10) (Keyboard,1,25)		address
1			(Mouse,2,10) (Keyboard,1,25)		address
1			(Mouse,2,10) (Keyboard,1,25)		address
1			(Mouse,2,10) (Keyboard,1,25)		address

Repeating Groups

✓ 1NF — Eliminate repeating groups

Orders_1NF					
order_id	cust_name	product_name	qty	price	address
1	Mouse		2	100	address
1	Mouse		2	120	address
1	Keyboard		1	125	address
1	Keyboard		1	125	address Ave
1	Mouse		2	100	address
1	Mouse		2	120	address
1	Keyboard		1	125	address
1	Keyboard		1	125	address Ave

No Repeating Groups (Atomic)

No Repeating Groups (Atomic)

Still Redundant, Partial Dependencies

✓ 2NF — Remove partial dependencies

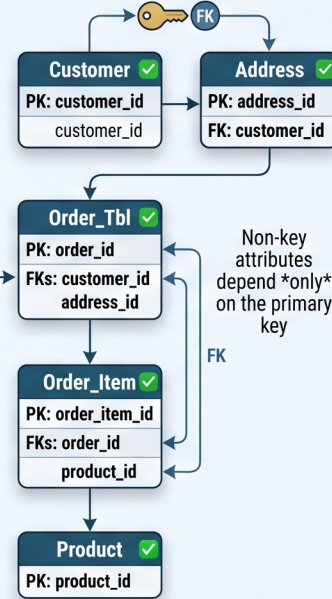
Order_Tbl (2NF)	
order_id PK	
cust_name	
cust_email	
address	

No Partial Dependency

Order_Item (2NF)			
order_id	product_name	qty	price
FK	product_name	2	100
	qty	1	125
	price	1	225

Still has Transitive Dependencies

✓ 3NF — Remove transitive dependencies



✓ 3NF (Satisfies BCNF)
- Final Standard Target

Why stop at 3NF?

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Eliminates anomalies,
Standard for OLTP systems,
Avoids excessive complexity,
Higher forms (4NF, 5NF)
not needed
(no multi-valued/join dependencies)

✓ 3NF